**Not Finished, But Abandoned**

by

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## ACKNOWLEDGMENTS

*It is customary for authors of academic books to include in their prefaces statements such as this: "I am indebted to ... for their invaluable help; however, any errors which remain are my sole responsibility." Occasionally an author will go further. Rather than say that if there are any mistakes then he is responsible for them, he will say that there will inevitably be some mistakes and he is responsible for them....*

*Although the shouldering of all responsibility is usually a social ritual, the admission that errors exist is not — it is often a sincere avowal of belief. But this appears to present a living and everyday example of a situation which philosophers have commonly dismissed as absurd; that it is sometimes rational to hold logically incompatible beliefs.*

— David C. Makinson (1965)

Above is the famous "preface paradox," which illustrates how to use the `wbepi` environment for epigraphs at the beginning of chapters. You probably also want to thank the Academy.

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## ABSTRACT

**FIXME: basically a placeholder; do not believe**

I did some research, read a bunch of papers, published a couple myself, (pick one):

1. ran some experiments and made some graphs,

2. proved some theorems

and now I have a job. I've assembled this document in the last couple of months so you will let me leave. Thanks!

# 1.1 What the heck is a plasma

## 1.1.1 Plasmas in the universe

## 1.1.2 How do plasmas affect us on earth?

## 1.1.3 Where does magnetic reconnection play into this whole shabang?

# 1.2 Magnetic Reconnection

## 1.2.1 The Generalized Ohm's Law

### 1.2.1.1 How do we derive this?

### 1.2.1.2 A breakdown of the terms

## 1.2.2 Regimes of Reconnection

### 1.2.2.1 Reconnection scaling parameters

i.e. why do we use $S$ and $L/d_i$ to characterize reconnection regimes?

Figure 1.1: Cartoon of places where reconnection can happen.

Figure 1.2: Diagram of reconnection regimes.

Figure 1.3: Diagram of Sweet-Parker reconnection.

Figure 1.4: Diagram of 2d hall reconnection

#### 1.2.2.2 Where do various systems line up?

#### 1.2.2.3 What are we interested in?

### 1.2.3 Sweet-Parker Reconnection

#### 1.2.3.1 A quick historical detour

#### 1.2.3.2 What are the assumptions that go into Sweet-Parker?

#### 1.2.3.3 Sweet-Parker Reconnection Rate Derivation

#### 1.2.3.4 Application to Real Systems

##### 1.2.3.4.1 What are some example parameters where reconnection occurs?

##### 1.2.3.4.2 What time scales does Sweet-Parker predict?

##### 1.2.3.4.3 What did we miss?

### 1.2.4 Hall Reconnection

#### 1.2.4.1 Why should the Hall term appear?

##### 1.2.4.1.1 What is the relevant physics that sets when the hall term matters

##### 1.2.4.1.2 In what kind of plasmas does the hall term appear and when does it not?

Figure 1.5: Flux tube changing size during reconnection causing pressure anisotropy build up.

Figure 1.6: Simulated kinetic reconnection with changing guide field.

### 1.2.5 Kinetic Reconnection

#### 1.2.5.1 What makes kinetic reconnection "kinetic"?

##### 1.2.5.1.1 What do collisions do?

###### 1.2.5.1.1.1 How quickly do collisions damp out non-Maxwellian features?

##### 1.2.5.1.2 What makes a system kinetic?

###### 1.2.5.1.2.1 What parameters set this?

###### 1.2.5.1.2.2 What should we do in the lab to get this?

#### 1.2.5.2 Where does Pressure Anisotropy come from?

##### 1.2.5.2.1 Movement of flux tube

##### 1.2.5.2.2 Effect of background field

###### 1.2.5.2.2.1 Too low of background field does...

###### 1.2.5.2.2.2 Too high of field does...

###### 1.2.5.2.2.3 Goldilocks!

Figure 1.7: Diagram from Ohia of with and without pressure anisotropy in simulation.

Figure 1.8: Some figure from Jan's papers thowing effects of pressure anisotropy

#### 1.2.5.3 How does pressure anisotropy change what the reconnection looks like

**1.2.5.3.1 Thinner reconnection layer?** Use results from Jan's more recent papers.

##### 1.2.5.3.2 Marginal Firehose Condition

##### 1.2.5.3.3 Concluding: what features should we look for in the data?

#### 1.2.5.4 Why do we want to measure it?

At the end of experimental setup I'll talk about why we want to do it in the lab but I think motivating why we care about kinetic reconnection as a process would be good here.

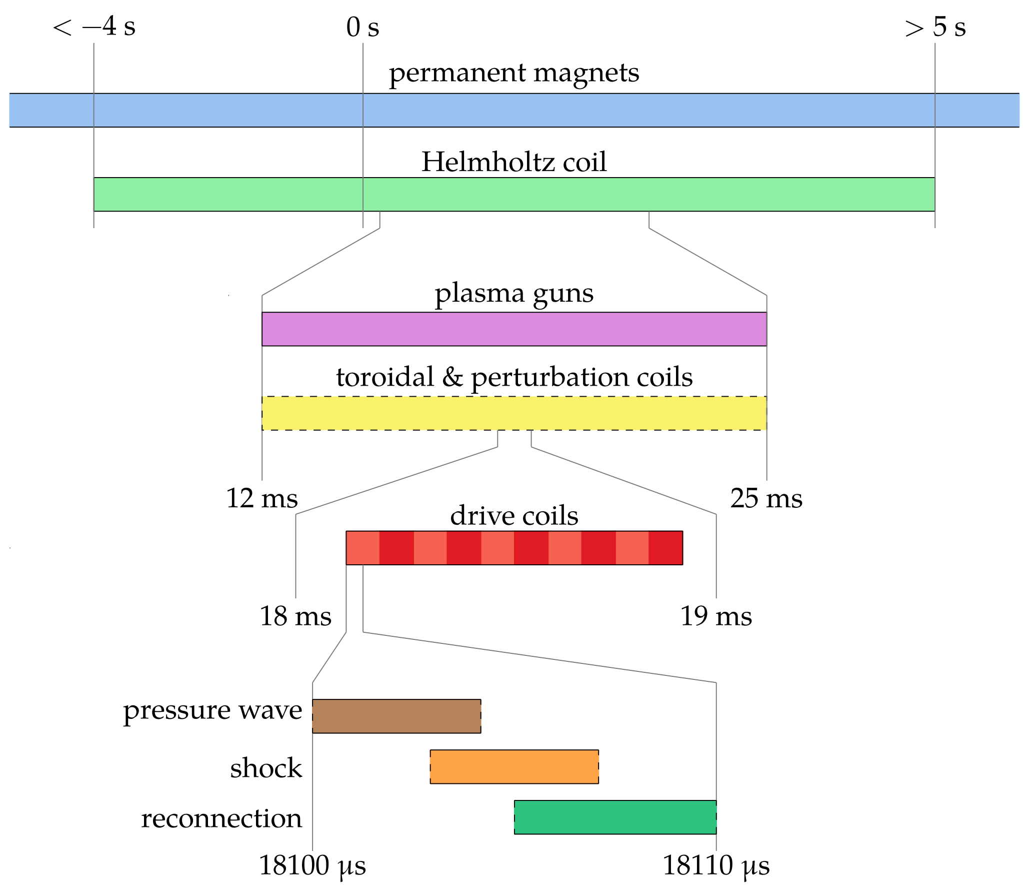


Figure 1.9: Timeline of experimental sequence.

Figure 1.10: Panels from flux arrays of experiment happening.

Figure 1.11: Circuit diagram of plasma gun.

Figure 1.12: Simplified cut of plasma gun insides.

## 1.3 Experimental Setup

### 1.3.1 Overall how does our experiment progress

### 1.3.2 Initial field

#### 1.3.2.1 Helmholtz

#### 1.3.2.2 Permanent magnets

#### 1.3.2.3 Toroidal

##### 1.3.2.3.1 Shifted coil

##### 1.3.2.3.2 Axial coil

#### 1.3.2.4 Perturbation coils

### 1.3.3 Plasma Generation

#### 1.3.3.1 How do our guns work

##### 1.3.3.1.1 Gas lines

##### 1.3.3.1.2 PFNs

##### 1.3.3.1.3 Puff trigger

##### 1.3.3.1.4 How to breakdown

Figure 1.13: Image from high speed camera of gun spewing

Table 1.1: List of diagnostics used in the experiment and what they can measure.

#### 1.3.3.2 How are the guns placed for injection?

##### 1.3.3.2.1 How much do they fill the machine

##### 1.3.3.2.2 What sits the filling fraction

#### 1.3.3.3 What goes wrong with them

##### 1.3.3.3.1 Burning up

**1.3.3.3.2 Burn holes in wall because of background field** Get image from Paul with hole in wall.

##### 1.3.3.3.3 Air cooling

##### 1.3.3.3.4 Optimizing gun operation

### 1.3.4 Drive field

#### 1.3.4.1 Individual coils

##### 1.3.4.1.1 Zone's between coils as plasma sources

#### 1.3.4.2 Drive Cylinder

Reference Paul's RSI + thesis.

### 1.3.5 Diagnostics Overview

#### 1.3.5.1 Bdot probes

##### 1.3.5.1.1 Basic principles

##### 1.3.5.1.2 Probe types

###### 1.3.5.1.2.1 Linear probe

###### 1.3.5.1.2.2 Hook probe

###### 1.3.5.1.2.3 Speed probe

###### 1.3.5.1.2.4 Flux array 1

###### 1.3.5.1.2.5 Flux array 2

**1.3.5.1.3 Data Interpretation** How do we convert the raw data into something useful?

#### 1.3.5.2 Hall probes

##### 1.3.5.2.1 Basic principles

##### 1.3.5.2.2 Issues and feasibility

#### 1.3.5.3 Langmuir probes

##### 1.3.5.3.1 Basic principles

##### 1.3.5.3.2 What do we need to do to make them work for our device?

##### 1.3.5.3.3 Look into XXX chapter

Table 1.2: Comparison of lab and space parameters.

##### 1.3.5.4 Fast camera

###### 1.3.5.4.1 What can we see with it?

###### 1.3.5.4.2 Potential use cases

### 1.3.6 How does the experiment compare to space

#### 1.3.6.1 Benefits of lab vs. space vs. simulation

##### 1.3.6.1.1 Simulation benefits and results

##### 1.3.6.1.2 Space benefits and results

##### 1.3.6.1.3 Lab benefits and results

### 1.3.7 Other lab reconnection experiments

#### 1.3.7.1 VTF, MRX, FLARE

#### 1.3.7.2 Exploding wire arrays

#### 1.3.7.3 Laser based

# 2 PRESSURE WAVE

## 2.1 Introduction

Once I get Jan's rewrite of the intro + conclusion I can figure out what might go here.

Figure 2.1: dBz/dt plots for different Helmholtz fields

Figure 2.2: Greess et al. 2022 paper showing simulations with pressure wave.

## 2.2 Observations that lead us to look at this wave

### 2.2.1 Speed probe measurements

#### 2.2.1.1 Flux expansion before shock layer

#### 2.2.1.2 Wavefront that depends on configuration

### 2.2.2 Simulations

#### 2.2.2.1 Different n and drive change feature

#### 2.2.2.2 Change reconnection layer post reflection

## 2.3 Physical understanding of wave

### 2.3.1 Initial plasma configuration

### 2.3.2 Release in magnetic pressure

#### 2.3.2.1 What would happen if plasma was immobile (replaced with perfect conductor)

All field is shielded at edge and there is no change in magnetic pressure in the bulk.

### 2.3.3 Wavefront tells plasma about pressure Release

#### 2.3.3.1 Analogous system

Sound wave.

How does a sound wave propogate in a plasma? Fast magnetosonic wave!

Figure 2.3: Simple diagram of sound wave in container

Table 2.1: Parameters for normalization

## 2.4 Wave Simulation

### 2.4.1 Approximations

### 2.4.2 Linearizing MHD

### 2.4.3 Plasma Equilibrium + Symmetries

#### 2.4.3.1 Types of Equilibrium

##### 2.4.3.1.1 theta-pinch

##### 2.4.3.1.2 screw-pinch

#### 2.4.3.2 Types of Symmetries

##### 2.4.3.2.1 z-symmetry

##### 2.4.3.2.2 toroidal-symmetry

### 2.4.4 Simplified 1D equation

#### 2.4.4.1 What does this look like?

A wave equation!

##### 2.4.4.2 Normalizations

##### 2.4.4.3 Normalized equation

### 2.4.5 Boundary conditions

#### 2.4.5.1 r = 0 Boundary

##### 2.4.5.1.1 v = 0 option

##### 2.4.5.1.2 xi = 0 option

##### 2.4.5.1.3 What do we choose?

#### 2.4.5.2 r = R Boundary

##### 2.4.5.2.1 What is physically happening at high radius in the coils?

##### 2.4.5.2.2 Loop voltage sets dBz/dt

##### 2.4.5.2.3 Assuming MHD so far so let's keep assuming!

##### 2.4.5.2.4 Vloop signal shape

### 2.4.6 Initial conditions

#### 2.4.6.1 What form can we expect plasma to take?

#### 2.4.6.2 What form do we use for these results

### 2.4.7 What does our simulation show?

#### 2.4.7.1 There is a wavefront traveling inwards at fast magnetosonic

##### 2.4.7.1.1 Low beta approximation giving Alfven

##### 2.4.7.2 Once wavefront passes, plasma moves out

##### 2.4.7.3 Converting to Bdot

##### 2.4.7.4 In Bdot we see flux moving outwards

As expected from MHD assumption.

## 2.5 How do we apply it to real data?

### 2.5.1 Base requirements

#### 2.5.1.1 Cylindrical symmetry

#### 2.5.1.2 Releasing magnetic pressure at high radius

#### 2.5.1.3 Measurements along a radial chord

#### 2.5.1.4 Known initial field

### 2.5.2 Extracing the wave speed from the data

#### 2.5.2.1 Wave tracking

##### 2.5.2.1.1 Edge finding

##### 2.5.2.1.2 Inertia

##### 2.5.2.1.3 Monte-carlo

##### 2.5.2.1.4 Centered finite difference

##### 2.5.2.1.5 Mean + Standard Deviation of Speed

Figure 2.4: Data vs. crude minimum density vs. empirical minimum density

Figure 2.5: Diagram of what the more complex model looks like.

Figure 2.6: CAD rendering of BRB w/ drive cylinder, flux probes, and Te probe.

### 2.5.3 Minimum measurable density

#### 2.5.3.1 Assuming sample resolution

##### 2.5.3.1.1 What does this assumption derive

##### 2.5.3.1.2 How does it compare to the data

##### 2.5.3.1.3 We need a better approximation

Figure 2.7: Plot of langmuir density and wave density

Figure 2.8: Diagram of sheath overlap for multitip langmuir probe

Figure 2.9: Plot of two-fluid dispresion relation

#### 2.5.3.2 Assuming sub-sample resolution

#### 2.5.3.3 A more complex model assuming the plasma is a step function * linear

## 2.6 Does it reproduce langmuir probe results

### 2.6.1 Experimental setup

#### 2.6.1.1 Drive cylinder

#### 2.6.1.2 Te probe along equator

### 2.6.2 Results

#### 2.6.2.1 Langmuir probe minimum measurable density

#### 2.6.2.2 Impacts of simple Langmuir model vs BRL

#### 2.6.2.3 What happens to the wave shape in simulation vs. experiment

##### 2.6.2.3.1 Two-fluid dispersion

##### 2.6.2.3.2 How might we expect this to change the wave

##### 2.6.2.3.3 How to simulate evolution approximately

##### 2.6.2.3.4 Dispersion simulation results

Figure 2.10: Results of simulating application of dispresion relation to pressure wave

Figure 2.11: Initial density before and after many shots

Figure 2.12: Initial density for varying gun number

## 2.7 Applications

### 2.7.1 Other machines where this is applicable

LAPD & MRX

### 2.7.2 Diagnosing gun failures

### 2.7.3 Measuring effect of different gun number and Helmholtz strength

#### 2.7.3.1 Gun number

#### 2.7.3.2 Helmholtz

### 2.7.4 Measuring diffusion coefficient

#### 2.7.4.1 Diffusion theoretical model

#### 2.7.4.2 Diffusion simulations

#### 2.7.4.3 Applications of pressure wave model

I think I need to do a bit more research here to show that we can actually extract a diffusion coefficient.

Figure 2.13: Initial density for varying Helmholtz

Figure 2.14: Diagram of flux mechanisms.

Figure 2.15: Example diffusion simulation

Figure 2.16: Example of cross-correlation

## 2.8 Future Research

### 2.8.1 Better wave tracking

#### 2.8.1.1 Cross-correlation

#### 2.8.1.2 Dynamic Time Warping

#### 2.8.1.3 Using a model of what the wave looks like

### 2.8.2 Using fast-magnetosonic speed instead of alfven

#### 2.8.2.1 Will this iteratively converge?

Figure 2.17: Example of DTW

# 3 MULTI-TIP LANGMUIR PROBE

Figure 3.1: Figure 2.6 from Joe's thesis

Figure 3.2: Picture of end of probe

Figure 3.3: Probe body from engineering sketch

Table 3.1: Length scales of probe

## 3.1 What's a Langmuir probe

## 3.2 Hardware

### 3.2.1 Shaft Components

#### 3.2.1.1 Garages

#### 3.2.1.2 Shaft materials

#### 3.2.1.3 Centering Collet

#### 3.2.1.4 Feed-thru

### 3.2.2 Tip Components

#### 3.2.2.1 Probe body

#### 3.2.2.2 Tip material + size

#### 3.2.2.3 Tip Shielding

### 3.2.3 Bdot Coils

## 3.3 Circuitry

### 3.3.1 Bias Source

#### 3.3.1.1 Manually Controlled

**3.3.1.1.1 Principles of operation** - Equation to use for bias calculation

Figure 3.4: Circuit diagram of old setup. Add switches.

Figure 3.5: Image of old setup

Figure 3.6: Circuit diagram of new setup. Add switches.

**3.3.1.1.2 Problems** - Had to be manually controlled for drastic bias changes - Wires often contacted each other or parts would lose contact

#### 3.3.1.2 Remote Controlled

**3.3.1.2.1 Principles of operation** - Equation to use for calculation

**3.3.1.2.2 Problems** - Destroyed a power supply - A resistor was designed and thus burnt out

Figure 3.7: Image of new setup

Table 3.2: Bias circuit component values

Table 3.3: Probe circuit component values

Table 3.4: Chen parameters

### 3.3.2 Probe Circuit

#### 3.3.2.1 Single-ended measurement

#### 3.3.2.2 Differential measurement

## 3.4 Plasma Theory

### 3.4.1 Simple Plasma Theory (Hutchinson)

#### 3.4.1.1 Derivation

#### 3.4.1.2 Pros & Cons

### 3.4.2 BRL

#### 3.4.2.1 Idea behind it

#### 3.4.2.2 Approximations of it

##### 3.4.2.2.1 Planar probe

##### 3.4.2.2.2 Chen Approximation

##### 3.4.2.2.3 Even more approximation

Figure 3.8: What does an inductive electric field do to the tip measurements diagram.

Figure 3.9: Comparison of expected E from Ohm's law and measured E.

Figure 3.10: Diagram of local minimums

### 3.4.3 Inductive electric fields

## 3.5 Curve Fitting

### 3.5.1 Least squares Fitting

#### 3.5.1.1 Benefits (easy)

#### 3.5.1.2 Issues (not representative of error)

### 3.5.2 ODR

#### 3.5.2.1 What does error look like in our signals

#### 3.5.2.2 How does ODR take that into account

### 3.5.3 Monte-carlo Fitting

### 3.5.4 Combining multiple shots

#### 3.5.4.1 Using some time points from each shot

#### 3.5.4.2 Combining all IV points and fitting that

## 3.6 Results

What is the benefit of doing all this work?

Figure 3.11: Example fit using many shots, multiple local parameter fits, and errors

### 3.6.1 Measurement of low densities

### 3.6.2 Noise resilience

### 3.6.3 Fits more accurate

- Each fit uses many data points
    - Don't have to arbitrarily cutoff data points

### 3.6.4 Extracting electric fields

# 4 PRESSURE ANISOTROPY PROBE

# 5 DIVERGENCE FREE INTERPOLATION? TALK TO JAN.

# 6 CONCLUSION

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